

SpatialAge Dusk

The chosen visual theme for the SpatialAge Vision Pro app, its dashboards and its pitch materials. Written so that a person or an AI coding model can apply it without guessing.



Source palette: #FBDAB4 peach · #D8AB8B tan · #A39193 mauve · #4E4B51 slate · #252322 charcoal

Instructions for AI models reading this file

You are working on SpatialAge, a visionOS (Apple Vision Pro) app. Apply the Dusk theme in this document to every UI, dashboard, slide or asset you create or change.

- Words in **MUST** / **NEVER** are hard rules. "Should" means the default; break it only with a stated reason.
- Take hex values from Section 2 exactly. Do not invent new colors. If a role is missing, derive it from the five source colors and say so in your change summary.
- Precedence when sources disagree: the user's instruction in chat, then this spec, then `specs/games/design.md` , then your own judgment.
- This spec replaces the old "Ink" look (dark navy sky `#0B0F17` with a floor grid). It does **not** change game signal colors, game rules, timings or scoring.
- Before finishing a task, run the checklist in Section 12.

Product in one paragraph

SpatialAge is a Vision Pro game built at Sundai Hack 143 (Biomarkers of Aging, Harvard, 4 October 2026). Players run short games that measure reaction, reach, balance and spatial memory with head and hand tracking. ScoreKit compares the results with published age norms and reports a **movement age** (code name `spatialAge` / `functional_age`). Two players can duel on two headsets. Pitch line: **"How old do you move?"**

Where things live in the repo

PATH	WHAT IT IS
<code>apps/vision/SpatialAge/Sources/Design/Theme.swift</code>	Color, size and motion tokens. Dusk goes here.
<code>apps/vision/SpatialAge/Sources/Design/MicroInteractions.swift</code>	Breathe, glow, pop, sink, snap. Keep as is.
<code>apps/vision/SpatialAge/Sources/Design/Tone.swift</code>	Procedural sound effects.
<code>apps/vision/SpatialAge/Sources/Stage/</code>	Immersive stage (sky dome, floor). Dusk environment goes here.
<code>apps/vision/SpatialAge/Sources/Catalog/</code>	Catalog and results windows (SwiftUI).
<code>apps/vision/SpatialAge/Sources/Games/</code>	8 games: Pendulum, Spark, Gate, Constellation, Orbit, Reach and Grab, Hole in the Wall, Color Dots.
<code>apps/dashboard/index.html</code>	Audience dashboard (static HTML, served by ingest).
<code>specs/games/design.md</code>	Existing design spec. Its signal colors and micro-interactions still apply.

The feel, in five words

Peaceful. Premium. Minimal. Warm. Legible. Clean, purposeful design by professionals, not beginners trying too hard. No "AI slop": no gradients for decoration, no emoji, no clutter, no stock flourishes.

Color tokens

Dusk is a **dark** theme: warm sunset sky, slate mountains, charcoal glass, cream type, peach accent.

UI tokens

TOKEN	VALUE	USE
bg	 #1B191A	Page / window background behind everything (dashboards, 2D)
surface	 #262324	Cards and boards on dashboards
surface2	 #2F2B2C	Nested cards, inactive tracks
fg	 #F7E9DA	Primary text on bg/surface
mute	 #A89698	Secondary text, labels, axis text
line	rgba(251,218,180,0.13)	Hairlines, table rules, grid lines
accent	 #FBDAB4	Primary button fill, key highlights
accentStrong	 #D8AB8B	Chart lines, progress fills, toggle on, focus ring
onAccent	 #252322	Text/icons on accent fills
accentSoft	rgba(216,171,139,0.20)	"Improved" chips, highlighted table rows, noise bands
accentGlow	rgba(251,218,180,0.45)	Glow around the primary button on gaze/hover
warn	 #FF9A62	Data-quality warnings only ("Tracking lost")

Glass tokens (windows and panels in space)

TOKEN	VALUE	USE
glass	rgba(37,35,34,0.52) + blur 28 px, saturate 1.5	Window and ornament background
glassStrong	rgba(37,35,34,0.80)	Window bar, dense panels, HUD on bright sky
glassEdge	rgba(251,218,180,0.22)	1 px border on glass
glassHighLight	rgba(251,218,180,0.18)	1 px inner top highlight
glassInk	 #FBEADA	All text and icons on glass
chip / chipStrong	rgba(251,218,180,0.09) / 0.20	Secondary buttons, cards inside windows / selected tab
gaze	rgba(251,218,180,0.22)	Hover highlight that follows the eye
shadow	rgba(0,0,0,0.55)	Window drop shadow

Game signal colors (fixed in every theme)

TOKEN	VALUE	MEANING
go	 #2F6BFF	Touch it
nogo	 #FF7A3D	Leave it
gold	 #FFC83D	Catch it / remember it
teal	 #14B8A6	Follow it

MUST keep these four exactly as in `Theme.swift`. Reaction times are only comparable across themes if stimuli look the same. **NEVER** use peach/tan accent colors, or any signal color, for decoration on in-world targets; peach next to `nogo` orange could be misread.

Environment and materials

The immersive background is a calm 3D nature scene at sunset. It replaces the Ink sky dome and floor grid.

Scene layers, back to front

LAYER	COLOR	NOTES
Sky, top	 #3A3240	Vertical gradient to horizon
Sky, horizon	 #FBD3A6	At about 62% of sky height
Sun + halo	 #FFE6C4	Low, near the horizon, off to one side (not behind targets)
Clouds	 #E9B9A0 at 45%	Soft, flat, slow drift
Far mountains	 #A39193	Followed by a haze band rgba(216,171,139,0.18)
Near mountains	 #6F6470	
Hills	 #4E4B51	
Lake	 #D8AB8B	Reflects the sky; faint light ripples
Trees	 #3A363C /  #252322	Silhouettes, two tones
Foreground grass	 #2D2A2C	Ground plane around the player
Ambient leaves	 #D8AB8B	A few drifting leaves, menus only

Rules

- **MUST** be life scale and world-locked so it moves in true 3D as the head moves (real parallax, never a flat skybox photo).
- **MUST** freeze all ambient motion (clouds, leaves, ripples) during scored trials. Any motion in view during play must be a game signal. This keeps the existing rule in `design.md`.
- Keep the play volume clear: nothing bright, moving or saturated within arm's reach (0.35–0.9 m) or behind where targets spawn.
- Mixed reality (passthrough) stays available for safety; the full Dusk scene is the immersive default.
- Hands stay visible: `upperLimbVisibility(.visible)`.

Glass windows in visionOS

- Use the system glass (`.glassBackgroundEffect()`) and tint it toward charcoal with an overlay of `glass` at 0.52. Corner radius about 46 pt for windows, full capsule for buttons and ornaments.
- 1 px `glassEdge` border plus 1 px top `glassHighlight`.
- Window bar (grab pill + close dot) stays the system one.
- Prefer one window plus one leading-edge tab ornament. No floating clutter.

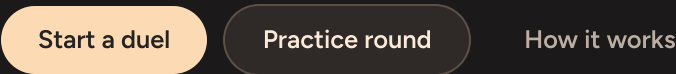
Typography and components

Type

ROLE	WEB / DASHBOARD / SLIDES	VISIONOS	USE
Hero number	Urbanist 200, -4% tracking	SF Pro Rounded, light	Movement age, last reaction time
Title	Urbanist 300-400	<code>.largeTitle</code> / <code>.title</code>	Screen and game titles
Body	Figtree 400	<code>.body</code> (SF Pro)	Instructions, one line each
Label	Figtree 600, uppercase, +16% tracking	<code>.caption.weight(.semibold)</code> , tracking 2-3	Eyebrows like "MOVEMENT AGE"
Data	JetBrains Mono / tabular figures	<code>.monospacedDigit()</code>	Participant codes, ms, cm

Numbers always use tabular figures. Big numbers are thin; labels are small and bold. Sentence case everywhere except labels.

Buttons



TYPE	LOOK	WHEN
Primary	Capsule, <code>accent</code> fill, <code>onAccent</code> text, weight 600	One per screen: the main next step
Secondary	Capsule, <code>chip</code> fill, <code>glassEdge</code> border, <code>glassInk</code> text	Alternatives ("Play one game", "Play again")
Tertiary	Text only, 72% opacity	Low-stakes ("Skip this game", "End duel")
Icon	Circle, <code>chip</code> fill, 1.35× icon	Back, close, play/pause

- **States:** rest; gaze (hover) = scale 1.035 and `accentGlow` halo on primary, `chipStrong` on secondary, plus a soft `gaze` spotlight following the eye (`.hoverEffect(.lift)` or `.highlight`); pinch = scale 0.95; disabled = 35% opacity, no glow.
- **MUST** give every target at least a 60 pt hit area and 16 pt spacing.
- Large buttons: primary actions on Home and Intro use the large size (about 60 pt tall).
- Layout rule from the brand brief: the one or two most useful actions sit in the center; everything else goes to the side or the tab ornament.

Other controls

- **Toggle:** track `chipStrong` , on = `accentStrong` , white knob.
- **Segmented control:** capsule track in `chip` ; selected segment `chipStrong` .
- **Chips:** "Improved" = `accentSoft` with a `accentStrong` dot; "Within noise" = neutral with a 40% dot; "Tracking lost" = `warn` dot.
- **Progress dots** (HUD): done = `glassInk` 75%, current = `gold` , upcoming = `chipStrong` .
- **Domain bars / XP bar:** 6 pt capsule, track `chipStrong` , fill `accentStrong` (XP: `accent` → `accentStrong`).
- **Tabs ornament:** vertical capsule at the window's leading edge: Home, Games, Progress, Duel. Labels appear on gaze.

Screens and copy

SCREEN	CONTENTS (TOP TO BOTTOM)
Home	Label "SPATIALAGE" · title "How old do you move?" · one line: "Eight short games read your reaction, reach, balance and memory. About six minutes, standing in place." · primary "Start a duel" · secondary "Play one game" · side card: last movement age, change since first visit chip, music mini-player.
Games	Title "Pick one, or play all eight" · primary "Play all" · grid of 8 cards (4×2): glyph, title, one-line instruction, uppercase "measures" label. Cards lift on gaze.
Game intro	Back icon · measures label · game title · one-line instruction · primary "Start" · toggle "Practice round first" (on by default) · side panel "Before you start" with 3 numbered steps · tertiary "Skip this game".
In play (HUD)	Small glass capsule at the top: game title, one instruction line, progress dots, skip (×). The window hides. NEVER show scores or numbers during play (exception: Pendulum's ruler).
Results	Label "MOVEMENT AGE" · hero number (one decimal) · chip "X years younger/older than [age]" · "80% interval A to B" · pace · five domain bars (Speed, Decision, Control, Memory, Consistency, 0–100) · per-game headline rows · secondary "Play again", primary "Next player".
Progress	Level name and XP bar · movement age trend with a shaded test-retest noise band · chips for which domains improved.
Duel	"Round N of 4" · two player panels (participant code, rounds won as dots) · primary "Start round N · [game]" · tertiary "End duel".

Game card copy (use exactly)

GAME	INSTRUCTION	MEASURES LABEL
Pendulum	Catch the weight when the cord lets go.	Catch latency and drop distance
Spark	Touch each light as it appears.	Reaction and reach
Gate	Touch blue. Leave orange.	Decision time and inhibition
Constellation	Watch the stars light, then touch them in order.	Spatial working memory
Orbit	Keep your fingertip inside the moving light.	Tracking error and lag
Reach and Grab	Keep your feet planted. Reach out and touch each object.	Reach distance and lean
Hole in the Wall	Fit your hands into the holes and hold still as the wall passes.	Sway and hand steadiness
Color Dots	Watch which dots light up, then touch only those.	Spatial recall and decisions

Voice

- Short, calm, second person. One instruction line per game. Active verbs on buttons ("Start", "Play again").
- A miss is quiet, never a failure message. No exclamation marks, no emoji.
- **MUST** say "movement age" and "game score". **NEVER** say biological age, diagnosis, or that playing prevents falls or lowers mortality. Allowed: "Your reaction time matches a typical 34-year-old's on this test."

Motion, sound and dashboards

Micro-interactions (unchanged from `MicroInteractions.swift`)

PRIMITIVE	BEHAVIOR
Appear	60 ms. Spawn time = first frame the target is in the scene.
Breathe	Live target scales 1.00 → 1.03 at 0.5 Hz.
Proximity glow	Emissive 0.6 + 2.4·c ² , c = closeness inside 30 cm.
Contact pop	90 ms scale 1 → 1.25 → 0; ring to 3× radius over 240 ms; tick pitched by reach speed.
Miss sink	250 ms desaturate, drop 3 cm, fade, soft low tone.
Release snap	Cord shortens and fades in 120 ms with a low tock.

UI motion: springs with low bounce (about 0.15), 200–350 ms. Respect Reduce Motion: drop ambient motion and breathe.

Sound

- **Music:** calm lofi loop made in Suno, 70–80 BPM, warm and slow for Dusk (working title "Golden Hour"). Loops forever without sounding repetitive: melodies and textures shift every few bars. Plays on menus; ducks to about 30% or stops during scored trials so audio never cues timing.
- **Effects:** soft sine partials with exponential decay (`Tone.swift`), pentatonic, spatialized from the object. Can be upgraded to produced Suno effects if they keep the same short, soft character.

Audience dashboard (`apps/dashboard`)

- Background `bg` , boards `surface` with 1 px `line` , radius 24 px.
- Header: pulsing "LIVE" dot (nogo color is fine here: it is 2D, not a stimulus), current game, "Round N of 4", Pause button.
- Two player panels side by side: participant code (mono), "Last catch" hero number in ms (Urbanist 200), sparkline of the last 16 reaction times on a fixed 250–750 ms scale with grid lines at 300/500/700 and an emphasized endpoint, median and caught count, rounds won as dots.
- Leaderboard by best catch: rank, code, best ms, movement age, gap. Current players highlighted with `accentSoft` .
- Event stats tiles: sessions, age range, valid-trial rate. Under 500 ms lag on local wifi.

Clinic view

- Patient code and age only, no names. Movement age line over visits in `accentStrong` , chronological age as a dashed `mute` line, test-retest noise band in `accentSoft` .
- Domain table with "+N improved" or "within noise" chips. Change only counts when it beats retest noise. Never a diagnosis.

Implementation

Swift: add to `Theme.swift`

```

/// Dusk theme. Spec: Dusk design spec v1.0. Signal colors (go, nogo, gold, teal) are unchanged.
enum Dusk {
    static func hex(_ v: UInt32, _ a: CGFloat = 1) -> UIColor {
        UIColor(red: CGFloat((v >> 16) & 0xFF) / 255, green: CGFloat((v >> 8) & 0xFF) / 255,
            blue: CGFloat(v & 0xFF) / 255, alpha: a)
    }
    // UI
    static let bg = hex(0x1B191A), surface = hex(0x262324), surface2 = hex(0x2F2B2C)
    static let fg = hex(0xF7E9DA), mute = hex(0xA89698), warn = hex(0xFF9A62)
    static let accent = hex(0xFBDAB4), accentStrong = hex(0xD8AB8B), onAccent = hex(0x252322)
    static let accentSoft = hex(0xD8AB8B, 0.20), accentGlow = hex(0xFBDAB4, 0.45)
    // Glass
    static let glass = hex(0x252322, 0.52), glassStrong = hex(0x252322, 0.80)
    static let glassEdge = hex(0xFBDAB4, 0.22), glassInk = hex(0xFBEADA)
    static let chip = hex(0xFBDAB4, 0.09), chipStrong = hex(0xFBDAB4, 0.20)
    // Environment
    static let skyTop = hex(0x3A3240), skyHorizon = hex(0xFBD3A6), sun = hex(0xFFE6C4)
    static let cloud = hex(0xE9B9A0), mountainFar = hex(0xA39193), mountainNear = hex(0x6F6470)
    static let hill = hex(0x4E4B51), lake = hex(0xD8AB8B), tree = hex(0x3A363C)
    static let treeDark = hex(0x252322), grass = hex(0x2D2A2C), haze = hex(0xD8AB8B, 0.18)
}

```

Machine-readable tokens

```

{
  "theme": "dusk", "version": "1.0", "mode": "dark",
  "source_palette": ["#FBDAB4", "#D8AB8B", "#A39193", "#4E4B51", "#252322"],
  "ui": {"bg": "#1B191A", "surface": "#262324", "surface2": "#2F2B2C", "fg": "#F7E9DA",
    "mute": "#A89698", "line": "rgba(251,218,180,0.13)", "accent": "#FBDAB4",
    "accentStrong": "#D8AB8B", "onAccent": "#252322", "accentSoft": "rgba(216,171,139,0.20)",
    "accentGlow": "rgba(251,218,180,0.45)", "warn": "#FF9A62"},
  "glass": {"fill": "rgba(37,35,34,0.52)", "fillStrong": "rgba(37,35,34,0.80)", "blurPx": 28,
    "edge": "rgba(251,218,180,0.22)", "highlight": "rgba(251,218,180,0.18)", "ink": "#FBEADA",
    "chip": "rgba(251,218,180,0.09)", "chipStrong": "rgba(251,218,180,0.20)",
    "gaze": "rgba(251,218,180,0.22)", "shadow": "rgba(0,0,0,0.55)"},
  "environment": {"skyTop": "#3A3240", "skyHorizon": "#FBD3A6", "sun": "#FFE6C4", "cloud": "#E9B9A0",
    "mountainFar": "#A39193", "mountainNear": "#6F6470", "hill": "#4E4B51", "lake": "#D8AB8B",
    "tree": "#3A363C", "treeDark": "#252322", "grass": "#2D2A2C", "leaf": "#D8AB8B",
    "haze": "rgba(216,171,139,0.18)"},
  "signals_fixed": {"go": "#2F6BFF", "nogo": "#FF7A3D", "gold": "#FFC83D", "teal": "#14B8A6"},
  "type": {"display": "Urbanist", "body": "Figtree", "mono": "JetBrains Mono", "visionOS": "SF Pro / SF Pro
Rounded"},
  "layout": {"minHitTargetPt": 60, "minSpacingPt": 16, "windowRadiusPt": 46, "buttons": "capsule"}
}

```

CSS (dashboards, web)

```

:root{--bg:#1B191A;--surface:#262324;--surface-2:#2F2B2C;--fg:#F7E9DA;--mute:#A89698;
--line:rgba(251,218,180,.13);--accent:#FBDAB4;--accent-strong:#D8AB8B;--on-accent:#252322;
--accent-soft:rgba(216,171,139,.2);--go:#2F6BFF;--nogo:#FF7A3D;--gold:#FFC83D;--follow:#14B8A6;color-
scheme:dark}

```

Do, don't, and the final checklist

Do

- Use only the tokens in this spec.
- One primary action per screen, centered.
- Thin hero numbers, small bold labels.
- Charcoal glass, cream text, peach for the main action.
- Freeze the scene during scored trials.
- Offer a practice round before every game.
- Keep instructions to one line.
- Label sample data as sample.

Don't

- Change go / nogo / gold / teal.
- Put peach, tan or any accent on in-world targets.
- Show scores during play.
- Use red for misses or harsh failure sounds.
- Add decorative gradients, emoji, glow on everything, or extra buttons in the center.
- Use bright white panels (that is the Meadow theme, not Dusk).
- Make health or biological-age claims.
- Store names. Participant codes only.

Checklist before you finish a task

1. Every color in the change maps to a token in Section 2 or 3.
2. The four signal colors are untouched and not reused for UI decoration.
3. Text on glass is `glassInk` ; text on peach is `onAccent` ; contrast is at least 4.5:1 for body text.
4. Hit targets are at least 60 pt with 16 pt spacing.
5. Each screen has at most one primary button.
6. No numbers on screen during play (Pendulum ruler excepted).
7. Ambient motion and music stop or duck during scored trials.
8. Copy uses "movement age", no medical claims, no exclamation marks.
9. Reduce Motion is respected.
10. If you touched the session schema, you bumped its version (repo rule).

Related files

- `PRD.md` : goals, team, claims wording, hack-day plan.
- `specs/games/*.md` : per-game rules and metrics. `specs/score.md` : ScoreKit and movement age.
- `concept/Branding.pdf` , `concept/Games Ideas (2).pdf` : Alex's original brand brief and game ideas.